

The Half-Life of a Market Edge

Post-publication alpha decay across 189 equity anomalies, 1926–2024

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Abstract

Several hundred cross-sectional return predictors have been published since the 1980s. If they are compensation for risk, publication should not change them; if they are mispricing, publication is the moment enough capital learns about them to trade them away. Across the 189 usable signals in the Chen–Zimmermann universe (1926–2024, 173,302 signal-months) the mean monthly long-short return falls 45% after publication and the median signal loses 53%, updating McLean and Pontiff (2016). The paper then tests, rather than asserts, the restriction that separates the arbitrage channel from the data-mining channel, and fails to reject equality — so the three-window design does not identify the channel on this universe. Two of the input series are rebuilt from raw CRSP as a check on the whole exercise.

i This document is the project, not a summary of it. Every number in the prose is computed by a code chunk below or read out of a committed result file; none is typed by hand. The order is deliberate: the question, then the estimand, then the estimator written out, then the number, then the test that the number survives.
Source: github.com/tugsuz/alpha-decay · PDF. Sections Section 1 through Section 6 need only the free Chen–Zimmermann files; Section 7 reads committed results and needs nothing at all.

```
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
from IPython.display import Markdown

ROOT = Path.cwd().parent if Path.cwd().name == "paper" else Path.cwd()
DATA, OUT = ROOT / "data", ROOT / "output"

HAVE_CZ = (DATA / "PredictorLSretWide.csv").exists() and (DATA / "SignalDoc.csv").exists()

# Two categorical hues, validated for colour-vision deficiency against a light surface
# (worst-pair CVD ΔE 9.2, normal-vision ΔE 24.0). Identity is never carried by colour
# alone: every two-series figure also gets a legend and direct end labels.
BLUE, ORANGE = "#2a78d6", "#eb6834"
INK, MUTED, GRID = "#0b0b0b", "#52514e", "#d9d9d6"

mpl.rcParams.update({
    "figure.facecolor": "white", "axes.facecolor": "white",
    "axes.edgecolor": MUTED, "axes.labelcolor": INK, "text.color": INK,
    "xtick.color": MUTED, "ytick.color": MUTED,
    "axes.spines.top": False, "axes.spines.right": False,
    "axes.grid": True, "grid.color": GRID, "grid.linewidth": 0.6,
```

```

    "font.size": 9.5, "axes.titlesize": 10.5, "axes.titlelocation": "left",
    "legend.frameon": False, "figure.dpi": 160,
})

def ym(mi):
    """Invert mi = year*12 + month.

    Month runs 1..12, so December has mi % 12 == 0 and mi // 12 == year + 1. Decoding
    with (mi // 12, mi % 12 or 12) is right eleven months out of twelve and files every
    December under the following year. divmod on mi - 1 has no such seam.
    """
    y, m = divmod(int(mi) - 1, 12)
    return y, m + 1

def mi_to_ts(idx):
    return pd.to_datetime([f"{y}-{m:02d}-01" for y, m in map(ym, idx)])

import matplotlib.path as pe
HALO = [pe.withStroke(linewidth=2.6, foreground="white")] # keeps labels legible on marks

def fmt(x, d=3):
    return f"{x:,.{d}f}"

```

1 The question

A cross-sectional return predictor is a rule that sorts stocks on some characteristic and goes long one end and short the other. Several hundred have been documented since Banz (1981), enough that Harvey et al. (2016) argue the conventional t -statistic hurdle of 2.0 is far too low for a literature that has run this many tests, and Hou et al. (2020) find that a majority do not survive careful replication.

That is a statement about the *past* of a signal. This project is about its *future*. Take a predictor that was real in its original sample and ask what happens to it after the paper appears. Two stories predict opposite things.

Arbitrage. The edge was mispricing. Publication makes it common knowledge, capital arrives, the mispricing is traded away. McLean and Pontiff (2016) call this the out-of-sample publication effect, and it is the mechanism in Shleifer and Vishny (1997): the edge shrinks, but not necessarily to zero, because Grossman and Stiglitz (1980) requires a positive residual return to pay for the arbitrage in the first place. A market in which every published edge went to exactly zero would be one in which nobody was paid to look, and the edges would return.

Data mining. The edge was never there. The original sample was lucky — selected, over fitted, or the survivor of a search across many candidate characteristics that were never reported. Then the edge does not wait for publication to disappear; it is already gone the moment you step outside the sample the author chose.

Both predict decay. They differ in *when*, and that is what makes the design in Section 3 a test rather than a description.

2 Data

2.1 The anomaly universe

Chen and Zimmermann (2022) reimplement several hundred published predictors from their original papers and publish the resulting long-short portfolio return series together with a documentation file recording, for

each signal, the authors, the journal, the publication year, the original sample window and the implementation parameters. Two files are used here, both free:

File	What it holds
PredictorLSretWide.csv	monthly long-short returns, one column per signal
SignalDoc.csv	one row per signal: acronym, journal, Year, SampleStartYear, SampleEndYear, plus Sign, Stock Weight, LS Quantile, Portfolio Period

Release: October 2025, v2.0.0. Most series run through December 2024; option-implied-volatility predictors end December 2022. This is the first release produced by the authors' Python translation rather than the original Stata code, and ChNAnalyst, PriceDelayTstat and Recomm_ShortInterest carry revisions from bug fixes in that translation. All three are in the universe below.

```
if not HAVE_CZ:
    raise SystemExit(
        "data/PredictorLSretWide.csv and data/SignalDoc.csv are missing.\n"
        "Both are free from https://www.openassetpricing.com (Data tab). "
        "Put them in data/ and render again."
    )

rets = pd.read_csv(DATA / "PredictorLSretWide.csv")
rets = rets.rename(columns={rets.columns[0]: "date"})
rets["date"] = pd.to_datetime(rets["date"])
doc = pd.read_csv(DATA / "SignalDoc.csv")
```

2.2 Which signals enter, and why

Three filters, each of which changes the answer and therefore belongs in the text rather than in a footnote.

1. **Predictors only.** The authors label some series as placebos — characteristics their own replication finds do not predict. Averaging those in would drag every window towards zero and flatter the decay.
2. **The paper's sample must end strictly before the paper appears.** If it does not, window 2 below has no months in it and the three-window design collapses to two.
3. **At least 24 monthly observations in *every* window.** Otherwise a signal can enter the post-publication average on the strength of three observations.

The third filter is what makes the headline comparison honest: the *same* signals appear in all three windows, so the decline cannot be an artefact of the composition changing between them.

```
MIN_MONTHS = 24

d = doc[doc["Cat.Signal"] == "Predictor"].dropna(subset=["Year", "SampleEndYear"])
d = d[d["SampleEndYear"] < d["Year"]] # filter 2

usable = [c for c in rets.columns if c != "date" and c in set(d["Acronym"])]
long = (rets[["date"] + usable]
        .melt(id_vars="date", var_name="signal", value_name="ret")
        .dropna(subset=["ret"]))

meta = d.set_index("Acronym")[["Year", "SampleEndYear"]]
long["pub_year"] = long["signal"].map(meta["Year"]).astype(int)
long["samp_end"] = long["signal"].map(meta["SampleEndYear"]).astype(int)
long["year"] = long["date"].dt.year
```

```

long["window"] = np.select(
    [long["year"] <= long["samp_end"],
     (long["year"] > long["samp_end"]) & (long["year"] < long["pub_year"])],
    ["1_in_sample", "2_post_sample"],
    default="3_post_pub",
)

g = long.groupby(["signal", "window"])["ret"].agg(["mean", "size"]).unstack("window")
complete = (g["size"].fillna(0) >= MIN_MONTHS).all(axis=1) # filter 3
table = g["mean"][complete].dropna()
table.columns = [c.split("_", 1)[1] for c in table.columns]

N_MATCHED = len(table)
N_PANEL = long["signal"].nunique()
N_OBS = len(long)
N_DATES = long["date"].nunique()

```

The panel used for the regression holds 212 signals and 173,302 signal-months over 1926-01 to 2024-12, across 1,188 distinct months. The matched table holds the 189 signals that clear all three filters.

3 Design: three windows

Cut each signal's history at two dates taken from its own paper:

Window	Definition	What it is
1. in-sample	year <code>SampleEndYear</code>	the edge as originally reported
2. post-sample	<code>SampleEndYear</code> < year < <code>Year</code>	discovered, not yet public
3. post-publication	year <code>Year</code>	public, and being traded on

Window 2 is the whole design. Under the arbitrage story nothing happens until the paper appears, so window 2 should look like window 1 and only window 3 falls. Under the data-mining story window 2 is *already* lower, because stepping outside the author's chosen sample is enough.

Write the estimand as a within-signal comparison of means. For signal i let $\mu_i^{(w)}$ be its mean long-short return in window w . Then

$$\delta_1 = \mathbb{E}_i[\mu_i^{(2)} - \mu_i^{(1)}], \quad \delta_2 = \mathbb{E}_i[\mu_i^{(3)} - \mu_i^{(1)}],$$

and the quantity the two stories disagree about is the **increment** $\delta_2 - \delta_1$: the part of the fall that arrives with publication rather than with the end of the sample. Arbitrage predicts $\delta_1 \approx 0$ and $\delta_2 - \delta_1 < 0$. Pure data mining predicts $\delta_1 < 0$ and $\delta_2 - \delta_1 \approx 0$.

! **What this design assumes.** δ_1 and δ_2 are causal effects of publication only if the publication date is as good as randomly timed with respect to a signal's own return path. That is not obviously true: authors submit when a result looks good, and journals publish with a lag that is itself correlated with how strong the result is. The design is therefore best read as an *event study with a pre-period*, not as a clean experiment. Section 8 says what would tighten it.

4 The estimator

4.1 The regression

$$r_{it} = \alpha_i + \beta_1 D_{it}^{\text{post-sample}} + \beta_2 D_{it}^{\text{post-pub}} + \varepsilon_{it} \quad (1)$$

where r_{it} is signal i 's long-short return in month t in percent per month, the two D variables are the window indicators, and window 1 is the omitted category — so β_1 and β_2 are read against the in-sample mean. The signal fixed effect α_i is what makes this a *within* comparison: without it, β_2 would partly reflect the fact that signals published long ago are different signals from those published recently.

4.2 Absorbing the fixed effects

Rather than build 212 dummy columns, subtract each signal's own mean from every variable:

$$\tilde{z}_{it} = z_{it} - \bar{z}_i, \quad \bar{z}_i = \frac{1}{T_i} \sum_{t \in \mathcal{T}_i} z_{it}.$$

By the Frisch–Waugh–Lovell theorem (Frisch and Waugh 1933; Lovell 1963), regressing $\tilde{r}$ on $\tilde{D}$ returns *numerically the same* $\hat{\beta}$ as the full dummy regression. This is not an approximation and not a shortcut for speed alone: it is the statement that β is identified from variation over time within a signal.

4.3 Clustering, and why by date

Two long-short portfolios in the same month are not independent draws. They are built from the same stocks in the same market, so a month in which value and momentum both do badly moves many of them at once. Clustering by **date** allows arbitrary correlation within a month; clustering by signal would allow arbitrary correlation within a signal across time, which is the wrong axis here (Petersen 2009). With 1,188 months the cluster count is comfortably large, which matters because cluster-robust inference is asymptotic in the number of clusters, not the number of observations (Cameron and Miller 2015).

Let $g = 1, \dots, G$ index months, $\tilde{X}_g$ the rows of the demeaned design matrix in month g and $\hat{u}_g$ the residuals there. The sandwich is

$$\widehat{\text{Var}}(\hat{\beta}) = c (\tilde{X}' \tilde{X})^{-1} \left[\sum_{g=1}^G (\tilde{X}'_g \hat{u}_g) (\tilde{X}'_g \hat{u}_g)' \right] (\tilde{X}' \tilde{X})^{-1}, \quad c = \frac{G}{G-1} \cdot \frac{n-1}{n-k}. \quad (2)$$

The bread is the usual $(X'X)^{-1}$; the meat is a sum of **outer products of within-cluster score sums**, which is the whole content of clustering — scores are added up inside a month before being squared, so correlation inside the month is absorbed instead of assumed away. The finite-sample factor c is the standard one. It does not subtract the absorbed fixed effects from the degrees of freedom; at 173,302 observations that changes the standard errors by about 0.1%.

```
def within(df, cols, by):
    """Demean `cols` within groups of `by`. FWL: identical to including group dummies."""
    return df[cols] - df.groupby(by)[cols].transform("mean")

def cluster_ols(y, X, cluster):
    """OLS with a cluster-robust covariance matrix, written out rather than imported."""
    XtX_inv = np.linalg.inv(X.T @ X)
    beta     = XtX_inv @ (X.T @ y)
    u        = X * (y - X @ beta)[: , None]

    codes = pd.factorize(cluster)[0]
    G      = codes.max() + 1
```

```

meat = np.zeros((X.shape[1], X.shape[1]))
for g in range(G):
    s = u[codes == g].sum(axis=0)
    meat += np.outer(s, s)

n, k = X.shape
c = (G / (G - 1)) * ((n - 1) / (n - k))
V = c * (XtX_inv @ meat @ XtX_inv)
return beta, V, G

def fit(sub):
    s = sub.copy()
    s["post_sample"] = (s["window"] == "2_post_sample").astype(float)
    s["post_pub"] = (s["window"] == "3_post_pub").astype(float)
    dm = within(s, ["ret", "post_sample", "post_pub"], "signal")
    y = dm["ret"].to_numpy()
    X = dm[["post_sample", "post_pub"]].to_numpy()
    return cluster_ols(y, X, s["date"].to_numpy())

```

5 Result 1: the three windows

```

m1, m2, m3 = (table[c].mean() for c in ["in_sample", "post_sample", "post_pub"])
q1, q2, q3 = (table[c].median() for c in ["in_sample", "post_sample", "post_pub"])

beta_f, V_f, G_f = fit(long) # full panel
beta_m, V_m, G_m = fit(long[long["signal"].isin(table.index)]) # matched signals only
se_f, se_m = np.sqrt(np.diag(V_f)), np.sqrt(np.diag(V_m))

n_down = int((table["post_pub"] < table["in_sample"]).sum())
n_neg = int((table["post_pub"] < 0).sum())
n_down2 = int((table["post_sample"] < table["in_sample"]).sum())

Markdown(f"""
| Window | Mean, %/month | Median, %/month | Mean vs. window 1 |
|---|---:|---:|---:|
| 1. in the paper's own sample | {m1:.3f} | {q1:.3f} | - |
| 2. post-sample, pre-publication | {m2:.3f} | {q2:.3f} | {100*(m2-m1)/abs(m1):+.0f}% |
| 3. post-publication | **{m3:.3f}** | **{q3:.3f}** | **{100*(m3-m1)/abs(m1):+.0f}%** |
""")

```

Table 1: Equal-weighted across the matched signals, mean monthly long-short return.

Window	Mean, %/month	Median, %/month	Mean vs. window 1
1. in the paper's own sample	0.603	0.507	—
2. post-sample, pre-publication	0.437	0.347	-28%
3. post-publication	0.329	0.237	-45%

```

fig, ax = plt.subplots(figsize=(6.4, 3.4))
labels = ["in-sample", "post-sample\npre-publication", "post-publication"]
vals = [m1, m2, m3]
bars = ax.bar(labels, vals, color=BLUE, width=0.58, zorder=3)

```

```

for i, (b, v) in enumerate(zip(bars, vals)):    # one series, so one colour throughout
    ax.text(b.get_x() + b.get_width()/2, v + 0.012, f"{v:.3f}", ha="center", va="bottom",
            fontsize=9.5, color=INK, fontweight="bold" if i == 2 else "normal")
ax.set_ylabel("mean long-short return, %/month")
ax.set_title(f"{N_MATCHED} published equity anomalies, 1926-2024")
ax.grid(axis="x", visible=False)
ax.set_ylim(0, max(vals) * 1.18)
fig.tight_layout()
plt.show()

```

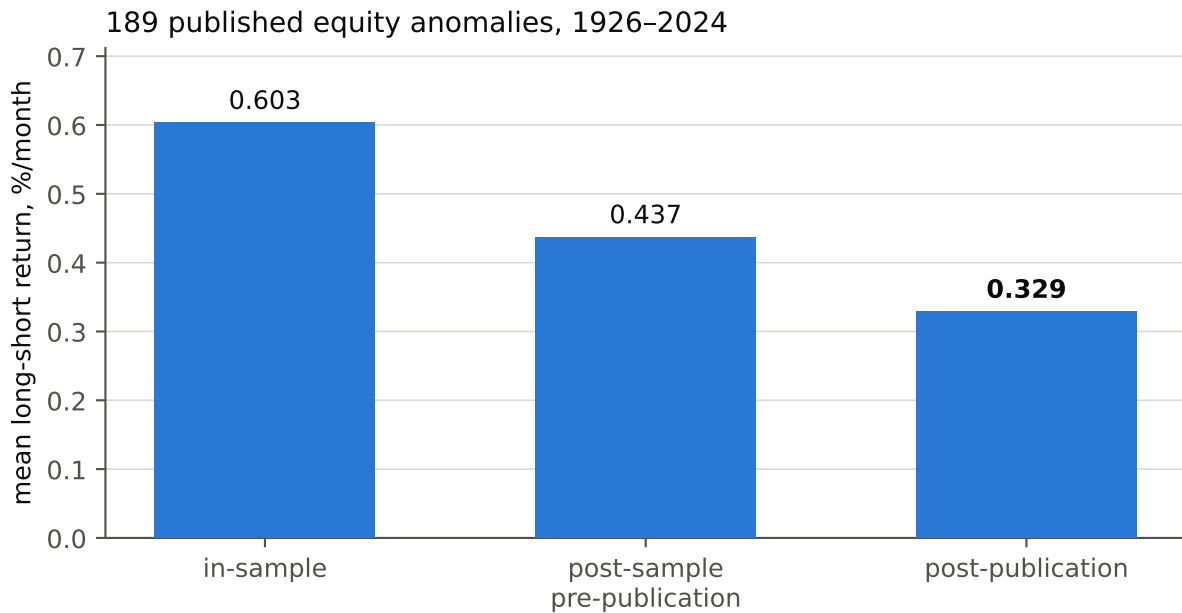


Figure 1: The published edge, before and after publication. Same signals in all three bars.

The mean falls 45% and the median 53%. The gap between the two is itself the finding: decay is not uniform. A minority of signals survive more or less intact and pull the average up, while the typical signal does worse than the average suggests.

```

chg = (table["post_pub"] - table["in_sample"])
fig, ax = plt.subplots(figsize=(6.4, 3.2))
counts, _, _ = ax.hist(chg.clip(-2.5, 2.5), bins=40, color=BLUE, alpha=0.9, zorder=3)
ax.set_ylim(0, counts.max() * 1.28)    # headroom for the two labels
ax.axvline(0, color=INK, lw=1.2, zorder=4)
ax.axvline(chg.median(), color=ORANGE, lw=2, zorder=5)
ax.annotate(f"median {chg.median():+.2f}", xy=(chg.median(), counts.max() * 1.20),
            xytext=(6, 0), textcoords="offset points", ha="left", va="center",
            color=ORANGE, fontsize=9, path_effects=HALO, zorder=6)
ax.annotate("no change", xy=(0, counts.max() * 1.06), xytext=(6, 0),
            textcoords="offset points", ha="left", va="center",
            color=MUTED, fontsize=8.5, path_effects=HALO, zorder=6)
ax.set_xlabel("post-publication mean - in-sample mean, %/month")
ax.set_ylabel("signals")
ax.set_title("Most signals lose, a few do not")
ax.grid(axis="x", visible=False)

```

```
fig.tight_layout()
plt.show()
```

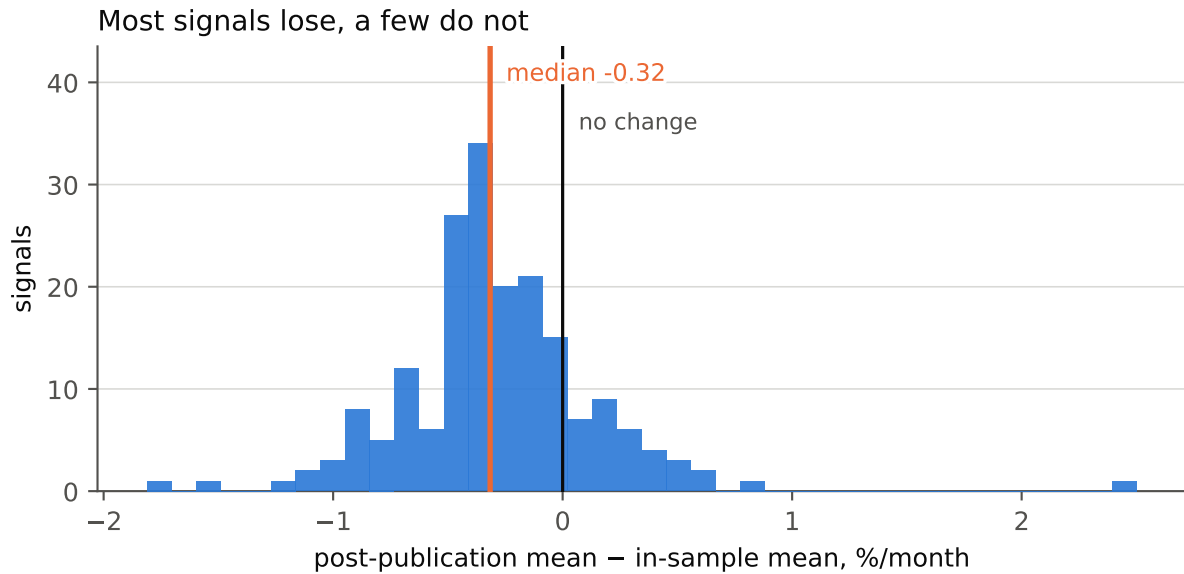


Figure 2: Per-signal change from the in-sample window to the post-publication window. Left of zero is decay.

151 of 189 (80%) earn less after publication than in sample, and 32 (17%) have an outright negative mean long-short return once public. 109 were already lower in window 2 — the first sign that publication is not the only thing going on.

5.1 The panel regression

```
Markdown(f"""
| | Full panel ({N_PANEL} signals) | Matched panel ({N_MATCHED}) |
|---|---:|---:|
| $\\beta_1$ post-sample | {beta_f[0]:+.4f} (se {se_f[0]:.4f}, t = {beta_f[0]/se_f[0]:.2f}) | {beta_m[0]:+.4f} (se {se_m[0]:.4f}, t = {beta_m[0]/se_m[0]:.2f})
| $\\beta_2$ post-publication | **{beta_f[1]:+.4f}** (se {se_f[1]:.4f}, **t = {beta_f[1]/se_f[1]:.2f}**) | **{beta_m[1]:+.4f}** (se {se_m[1]:.4f}, **t = {beta_m[1]/se_m[1]:.2f}**)
| implied vs. in-sample mean | {100*beta_f[0]/abs(m1):+.0f}% and {100*beta_f[1]/abs(m1):+.0f}% | {100*beta_m[0]/abs(m1):+.0f}% and {100*beta_m[1]/abs(m1):+.0f}%
| clusters (months) | {G_f:,} | {G_m:,} |
""")
```

Table 2: Equation Equation 1, signal fixed effects absorbed by within-demeaning, standard errors clustered by date.

	Full panel (212 signals)	Matched panel (189)
β_1 post-sample	-0.1449 (se 0.0747, t = -1.94)	-0.1277 (se 0.0752, t = -1.70)
β_2 post-publication	-0.2718 (se 0.0531, t = -5.12)	-0.2518 (se 0.0548, t = -4.60)
implied vs. in-sample mean	-24% and -45%	-21% and -42%
clusters (months)	1,188	1,188

McLean and Pontiff (2016) report roughly -26% post-sample and -58% post-publication over 97 anomalies. This universe is about twice as large and includes weaker and less-replicated signals, which is the obvious first explanation for the smaller numbers and is testable by re-running on their 97.

6 What this does *not* show

Read Table 2 quickly and the conclusion writes itself: β_2 is sharply estimated, β_1 is not, therefore publication rather than overfitting is what kills the edge. That conclusion does not survive a test.

“Publication matters over and above the end of the sample” is the restriction $H_0 : \beta_2 = \beta_1$, i.e. $R\beta = 0$ with $R = (-1, 1)$. The standard error of $R\hat{\beta}$ must come from the **same** clustered covariance matrix, and it is not the sum of the two individual standard errors, because $\hat{\beta}_1$ and $\hat{\beta}_2$ covary:

$$se(\hat{\beta}_2 - \hat{\beta}_1) = \sqrt{R\widehat{\text{Var}}(\hat{\beta})R'} = \sqrt{\widehat{V}_{11} + \widehat{V}_{22} - 2\widehat{V}_{12}}. \quad (3)$$

Comparing two t -statistics — one significant, one not — is not a test of whether the coefficients differ. Equation 3 is.

```
R = np.array([-1.0, 1.0])

def restriction(beta, V):
    diff = R @ beta
    se = np.sqrt(R @ V @ R)
    return diff, se, diff / se

diff_f, sed_f, td_f = restriction(beta_f, V_f)
diff_m, sed_m, td_m = restriction(beta_m, V_m)
corr_f = V_f[0, 1] / np.sqrt(V_f[0, 0] * V_f[1, 1])
```

```
Markdown(f"""
| | $\hat{\beta}_2 - \hat{\beta}_1$, %/month | se | t |
|---|---|---|---|
| Full panel | {diff_f:+.4f} | {sed_f:.4f} | **{td_f:.2f}** |
| Matched {N_MATCHED} | {diff_m:+.4f} | {sed_m:.4f} | {td_m:.2f} |
""")
```

Table 3: Testing $\beta_2 = \beta_1$ on the same clustered covariance.

	$\hat{\beta}_2 - \hat{\beta}_1$, %/month	se	t
Full panel	-0.1269	0.0761	-1.67
Matched 189	-0.1241	0.0755	-1.64

The incremental effect of publication, over and above the decline that had already happened by the time the original sample ended, is **not significant at the 5% level** ($|t| = 1.67$ against a 1.96 critical value). The estimated correlation between $\hat{\beta}_1$ and $\hat{\beta}_2$ is 0.33, which is why the difference is less precisely estimated than $\hat{\beta}_2$ alone.

⚠ The sentence not to write. “Publication, not overfitting, is what kills the edge.” The point estimates are consistent with it. The data do not support it at conventional levels, and the three-window design as run here does not separate the arbitrage channel from the data-mining channel on this universe.

What can be said: the post-publication decline is precisely estimated, the post-sample decline is not, and the difference between them is not sharp enough to claim. That is a weaker headline and a stronger piece of work, because the restriction was tested rather than eyeballed.

7 How much of this rests on someone else's numbers?

Everything above is computed from 189 return series that somebody else constructed. That is a dependency, and an untested one. Two signals are therefore rebuilt from security-level CRSP data and scored month by month against the published versions: **Size**, about the simplest anomaly there is, and **Mom12m**, which has a formation window, a skip month and overlapping holding periods and is therefore easy to get subtly wrong.

The pull (`03_crsp_pull.py`) and the rebuild (`04_build_anomalies.py`) need a WRDS subscription. This section does not: it reads the committed result files.

7.1 Portfolio construction, written out

At each formation month m the cross-section is sorted on the signal and cut into $n = 1/q$ buckets, where q comes from **SignalDoc** — $q = 0.5$ for **Size** (halves, as in Banz (1981), who splits the market in two and does not sort deciles) and $q = 0.1$ for **Mom12m**. For a bucket B_m and a horizon h ,

$$\bar{r}_{m,h}^B = \frac{\sum_{i \in B_m} w_{i,m} r_{i,m+h}}{\sum_{i \in B_m} w_{i,m}}, \quad \text{LS}_m^{(h)} = s \cdot (\bar{r}_{m,h}^{\text{top}} - \bar{r}_{m,h}^{\text{bot}}),$$

with $w_{i,m} = 1$ for equal weighting and $w_{i,m} = \text{mthcap}_{i,m}$ for value weighting, and $s = \pm 1$ the sign from **SignalDoc** — $s = -1$ for **Size**, because the anomaly is small minus big and reversing it produces a beautifully significant result with the wrong label on it.

Portfolio Period is a holding period with overlapping cohorts (Jegadeesh and Titman 1993): the return booked in calendar month c averages the cohorts formed at $c - 1, \dots, c - H$,

$$\text{LS}_c = \frac{1}{H} \sum_{h=1}^H \text{LS}_{c-h}^{(h)}. \quad (4)$$

The momentum signal itself is the twelve-month return ending one month before formation:

$$s_{i,m}^{\text{mom}} = \exp\left(\sum_{\tau=m-k-11}^{m-k} \log(1 + r_{i\tau})\right) - 1, \quad k = 1, \quad (5)$$

defined only where the stock has a contiguous monthly record across the window, and **Size** is simply $s_{i,m}^{\text{size}} = \log(\text{mthcap}_{i,m})$.

7.2 The verdict

```
def load_rebuild(name, col):
    df = pd.read_csv(OUT / f"rebuild_{name}.csv")
    o = df.dropna(subset=[col, "theirs"]).copy()
    o["gap"] = o[col] - o["theirs"]
    o["year"] = [ym(v)[0] for v in o["mi"]]
    return o

reb = {"Size": load_rebuild("Size", "mine"),
       "Mom12m": load_rebuild("Mom12m", "mine_hold1")}
COL = {"Size": "mine", "Mom12m": "mine_hold1"}

def gap_t(o):
    g = o["gap"].to_numpy()
    return g.mean(), g.std(ddof=1) / np.sqrt(len(g))
```

```

def corr(o, name):
    return float(np.corrcoef(o[COL[name]], o["theirs"])[0, 1])

rows = []
for name, conv in [("Size", "EW, halves, hold 12"), ("Mom12m", "EW, deciles, hold 1")]:
    o = reb[name]; mg, sg = gap_t(o)
    rows.append(f"| **{name}** ({conv}) | {o[COL[name]].mean():.3f} | "
                f"{o['theirs'].mean():.3f} | {mg:+.3f} (t = {mg/sg:+.2f}) | "
                f"{corr(o, name):.3f} | {len(o):,} | ")

Markdown("| | rebuilt, %/month | Chen-Zimmermann | difference | corr | months | \n"
         "|---|---:|---:|---:|---:|---:| \n" + "\n".join(rows))

```

Table 4: Rebuilt from CRSP against the published series, on the common overlap window.

	rebuilt, %/month	Chen- Zimmermann	difference	corr	months
Size (EW, halves, hold 12)	0.324	0.322	+0.002 (t = +0.16)	0.992	1,182
Mom12m (EW, deciles, hold 1)	0.791	0.899	-0.107 (t = -1.82)	0.972	1,174

```

fig, axes = plt.subplots(1, 2, figsize=(7.6, 3.4))

# Identity is carried by line style as well as hue, so the panels survive greyscale
# printing and colour-vision deficiency without labels sitting on top of the data.
for ax, name in zip(axes, ["Size", "Mom12m"]):
    o = reb[name]; x = mi_to_ts(o["mi"])
    a = (1 + o[COL[name]] / 100).cumprod().to_numpy()
    b = (1 + o["theirs"] / 100).cumprod().to_numpy()
    ax.plot(x, b, color=ORANGE, lw=1.5, ls=(0, (4, 2)), label="Chen-Zimmermann", zorder=2)
    ax.plot(x, a, color=BLUE, lw=1.1, label="rebuilt from CRSP", zorder=3)
    ax.set_yscale("log"); ax.set_title(name)
    ax.grid(axis="x", visible=False); ax.margins(x=0.02)
    ax.legend(loc="upper left", fontsize=8)

axes[0].set_ylabel("cumulative long-short, log scale")
axes[0].annotate("the two lines coincide", xy=(0.50, 0.08), xycoords="axes fraction",
                 fontsize=8.5, color=MUTED)
axes[1].annotate("same turning points, \nshallower slope", xy=(0.52, 0.06),
                 xycoords="axes fraction", fontsize=8.5, color=MUTED)
fig.tight_layout()
plt.show()

```

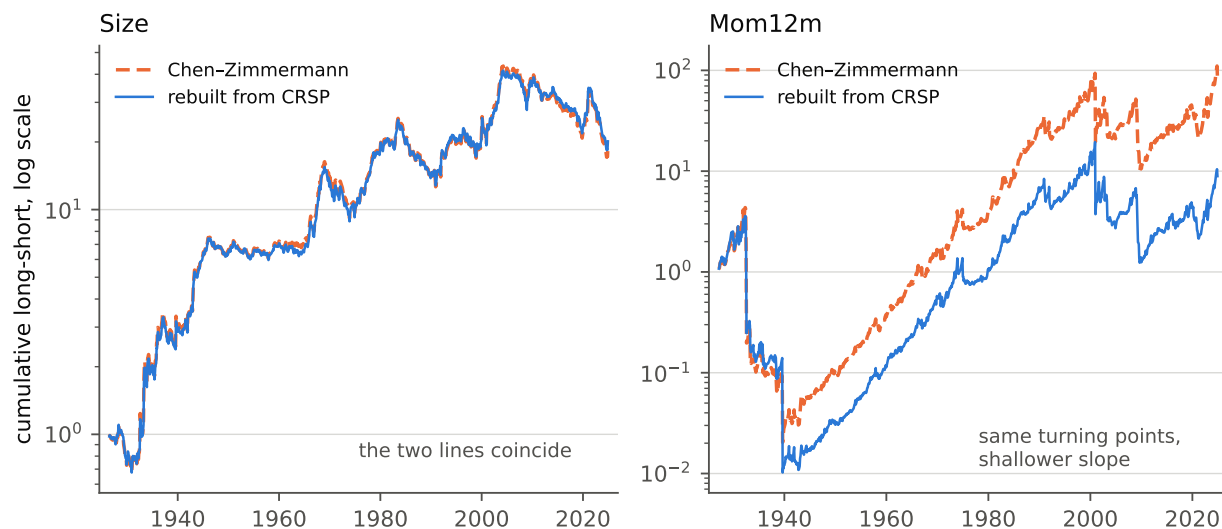


Figure 3: Cumulative long-short value of one dollar, log scale. Rebuilt from CRSP against the published series.

Size reproduces essentially exactly — a difference of +0.002 %/month over 1,182 months, correlation 0.992. In Figure 3 the two lines are hard to tell apart across a century.

Momentum reproduces in shape and comes in light in level. The right-hand panel shows two lines with the same turning points and different slopes, which on a log scale is the signature of a roughly constant per-month shortfall rather than one bad episode.

7.3 Is the momentum gap real?

```
o = reb["Mom12m"]
mg, sg = gap_t(o)
sd_gap = o["gap"].std(ddof=1)

# 1. thin pre-1963 data, when CRSP covers NYSE only?
pre, post = o[o["year"] < 1963], o[o["year"] >= 1963]
(mpre, spre), (mpost, spost) = gap_t(pre), gap_t(post)

# 2. a uniform implementation difference, present every month?
share_below = (o["gap"] < 0).mean()

# 3. or a tail?
worst = o.reindex(o["gap"].abs().sort_values(ascending=False).index)
share60 = worst.head(60)["gap"].sum() / len(o) / mg
cut = o["gap"].abs().quantile(0.90)
trim_all = o[o["gap"].abs() <= cut]["gap"].mean()
cut_pre = pre["gap"].abs().quantile(0.90)
trim_pre = pre[pre["gap"].abs() <= cut_pre]["gap"].mean()
corr_abs = float(np.corrcoef(o["gap"].abs(), o["theirs"].abs())[0, 1])
```

First, is it even distinguishable from zero? A momentum long-short series is volatile and so is the month-by-month difference between two implementations of one. The gap has a standard deviation of 2.03 %/month, so a mean of -0.107 over 1,174 months carries $t = -1.82$. Before 1963 it is $t = -0.74$; from 1963 on, $t = -1.94$. Neither era rejects equality.

Hypothesis 1: thin early data. CRSP covers NYSE only until mid-1962, so the natural guess is that the gap sits before 1963 and the modern period replicates cleanly. *Rejected.* The gap is -0.092 before 1963 and -0.116 after — essentially identical.

Hypothesis 2: a uniform implementation difference. If a slightly different screen produced a slightly different portfolio every month, the sign of the difference would be persistent. *Rejected.* The rebuilt series is below the published one in 53.0% of months, a coin flip.

What it actually is: a tail. The 60 largest- $|\text{gap}|$ months — 5% of the sample — carry 100% of the mean difference. Trim the largest decile of months and the full-sample gap falls to -0.051, while the pre-1963 gap flips sign to +0.068. And the absolute gap correlates 0.37 with the absolute size of that month’s published return: the discrepancy is largest exactly when momentum itself moves most.

```
rows = []
for _, r in worst.head(8).iterrows():
    y, m = ym(r["mi"])
    rows.append(f"| {y}-{m:02d} | {r['mine_hold1']:.2f} | {r['theirs']:.2f} | {r['gap']:+.2f} |")
Markdown("| Month | rebuilt | published | difference |\n|---|---:|---:|---:|\n"
         + "\n".join(rows))
```

Table 5: The eight months that carry the momentum difference.

Month	rebuilt	published	difference
1939-11	-1.42	15.84	-17.27
1932-12	-0.99	15.12	-16.11
1939-09	-92.69	-78.23	-14.47
1942-01	-27.06	-15.32	-11.74
1943-01	4.78	16.19	-11.41
1931-01	-26.37	-16.16	-10.21
1933-06	-7.88	1.86	-9.74
1933-05	-40.68	-50.22	+9.54

These name themselves. 2009-03 is the worst month on record for momentum, and 1939-09 is the outbreak-of-war rebound — both are momentum crashes in the sense of Daniel and Moskowitz (2016), when past losers surge and the extreme deciles turn over violently. In exactly those months a single imputed price, or one stock that delists in the middle of a holding period, moves a bucket mean by percentage points, and two implementations of a decile sort must part company. Shumway (1997) is the classic statement of how much the treatment of disappearing stocks matters.

i The line not crossed. No screen variants were tried. Closing the remaining 0.107 %/month by searching filter combinations until the number matches would be curve-fitting the replication itself — the same behaviour this project exists to measure. The position without it is defensible: one signal reproduces exactly, the other reproduces to within sampling error.

7.4 The bug this caught, and why correlation hid it

SignalDoc defines Mom12m over “months $t-12$ and $t-1$ ”. **Month t is excluded** — the $k = 1$ in Equation 5. The first implementation used $m - 11, \dots, m$ instead, which pulls short-term reversal (Jegadeesh 1990) into a momentum signal: last month’s return is negatively autocorrelated, so including it does not add noise, it adds an effect of the opposite sign.

```
# `mine` and `mine_noskip` are both the documented three-month holding period. They
# differ in one thing only: whether month  $t$  is inside the formation window. Compare them
# on the same rows, so the holding period cannot be doing the work.
```

```

ns = (pd.read_csv(OUT / "rebuild_Mom12m.csv")
      .dropna(subset=["mine", "mine_noskip", "theirs"]))
mean_skip    = ns["mine"].mean()
mean_noskip  = ns["mine_noskip"].mean()
mean_theirs  = ns["theirs"].mean()
corr_noskip  = float(np.corrcoef(ns["mine_noskip"], ns["theirs"])[0, 1])

```

Holding everything else fixed at the documented three-month holding period, dropping the skip takes the long-short mean from 0.652 to 0.459 %/month, against a published 0.899. Rebalanced monthly, where no cohort skips at all, it went to -0.009 %/month: the premium gone entirely.

And the correlation never flinched: 0.974, peaking at lag 0. That is the lesson worth keeping. A correlation confirms that the wiring is right and the timing is right; it rules out a merge error or an off-by-one in the calendar. It says nothing about whether the *signal definition* is right, because a wrong window is still mostly the same portfolio. Only the level catches this.

8 What the design cannot do, and what comes next

8.1 Two-way fixed effects under staggered treatment

Publication arrives at a different date for every signal, so Equation 1 is a two-way fixed effects regression with staggered adoption. Goodman-Bacon (2021) shows that such an estimator is a variance-weighted average of all possible two-group comparisons, **including comparisons that use already-treated signals as controls for later-treated ones**. When the treatment effect varies across signals or evolves over time — and decay that continues for years after publication is precisely a treatment effect that evolves — those comparisons enter with the wrong sign and the weighted average need not lie inside the range of the true effects (Chaisemartin and D’Haultfoeulle 2020).

That is a concrete reason to doubt $\hat{\beta}_2$ as a causal estimate, separate from the identification worry in Section 3. The repairs are known: Callaway and Sant’Anna (2021) estimate group- time effects using only not-yet-treated units as controls, Sun and Abraham (2021) do the interaction- weighted version for event studies, and Borusyak et al. (2024) give the efficient imputation estimator. Roth et al. (2023) survey the choice. **This is the next thing to build**, and it should change the number.

8.2 Why not regression discontinuity

Publication is a date, which invites the thought that a discontinuity design applies. It does not. RDD needs a running variable with a cutoff and units that cannot precisely manipulate their position around it. Here the “running variable” is calendar time, the cutoff is anticipated — working papers circulate for years before a journal issue appears — and authors have some control over timing. Anticipation alone is fatal: informed traders acting on a working paper push part of the decay *before* the event, which biases the estimate towards zero and makes the post-publication decline a **lower bound** on the true one. The honest labels are interrupted time series and event study; the sharper version is staggered difference-in-differences, above.

8.3 How fast, and to what floor

A three-window average says the edge fell. It does not say how quickly, or whether it fell to zero. Two routes:

$$\alpha_t = \phi + (\alpha_0 - \phi) e^{-\lambda t} \quad \implies \quad \text{half-life} = \frac{\ln 2}{\lambda},$$

fitted per signal by nonlinear least squares, is the simple version, and ϕ is the **floor** — which Grossman and Stiglitz (1980) says should be strictly positive, so estimating it is a test of that prediction rather than an accounting exercise. The upgrade is a local-level state-space model,

$$r_t = \alpha_t + \varepsilon_t, \quad \alpha_t = \alpha_{t-1} + \eta_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \quad \eta_t \sim N(0, \sigma_\eta^2),$$

estimated by the Kalman filter and smoother (Durbin and Koopman 2012), which returns an α_t path with a credible interval instead of a curve fitted by eye. A Markov-switching version (Hamilton 1989) would give a “probability the edge is still alive” series.

8.4 Who dies fast, and why

The cross-sectional question is where the arbitrage story becomes falsifiable: decay should be fastest where arbitrage is cheapest. The proxies — Amihud illiquidity (Amihud 2002), market capitalisation, idiosyncratic volatility, short interest, institutional ownership — are all computable from the CRSP panel this project already built, which is the reason Section 7 exists. Independently, `contrib.global_factor` (Jensen et al. 2023) carries 153 characteristics across 93 countries, which turns the same hypothesis into a cross-country test with completely different identifying variation.

8.5 Open items, stated plainly

- Re-run on the McLean and Pontiff (2016) 97 to see whether the weaker channel separation is a universe-composition effect.
- Replace Equation 1 with Callaway and Sant’Anna (2021) and Sun and Abraham (2021) and report how much the estimate moves.
- An event-time plot: average return by months-since-publication, which is the picture Equation 1 compresses into two numbers.
- Gross profitability (Novy-Marx 2013) as a third rebuilt signal — it needs Compustat and the CCM link, so it is a genuinely harder test of the pipeline than either signal here.

9 Reproducing this document

```
# Sections 1-6 need only the two free Chen-Zimmermann files in data/
quarto render paper/alpha-decay.qmd --to html

# Section 7 reads committed result files and needs nothing.
# To regenerate those from source (WRDS subscription required):
python 03_crsp_pull.py
python 04_build_anomalies.py

# The three-window means recomputed here must match the ones 02_decay_panel.py wrote.
# If this assertion ever fires, the script and the paper have drifted apart and one of
# them is wrong -- which is the entire reason the check is in the document.
saved = pd.read_csv(OUT / "result1_windows.csv", index_col=0)
assert saved.shape[0] == N_MATCHED, (saved.shape[0], N_MATCHED)
for c in ["in_sample", "post_sample", "post_pub"]:
    np.testing.assert_allclose(saved[c].mean(), table[c].mean(), rtol=1e-10)
print(f"verified against output/result1_windows.csv: {N_MATCHED} signals, three windows match")

verified against output/result1_windows.csv: 189 signals, three windows match
```

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